Yukai Wang

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Education

Shanghai Jiao Tong University — Shanghai, China
Bachelor's Degree in Electrical and Computer Engineering GPA: 3.78/4.00 — *Aug. 2023 – Aug. 2027 (expected)*

University of Michigan — Ann Arbor, US
Bachelor's Degree in Industrial and Operations Engineering GPA: 4.00/4.00 — *Aug. 2025 – May 2027 (expected)*

Research Experience

Undergraduate Research Assistant – LLM Agent Failure Monitoring — Jan. 2026 – Present
University of Michigan | Code — *Advisor: Prof. Raed Al Kontar*

- Built a monitoring and tree search framework around mini-SWE-agent that logs 17 step-level signals and runs official SWE-bench evaluation on HPC with Slurm and Singularity.
- Designed a recursive binary rollout tree that forks the agent's sandbox every K steps to estimate each intermediate state's probability of eventual success, generating about 150K labeled trajectory prefixes from 707 SWE-bench tasks.
- Trained a calibrated prefix-level failure monitor on 148 features (test AUROC 0.89), flagging 84% of failing trajectories on average with 38% of the trajectory still remaining.
- Benchmarked linear, LightGBM, MLP, and XGBoost scorers with sequential alarm rules (CUSUM, EWMA, consecutive-K). LightGBM with tuned CUSUM detected 86% of failing trajectories at a 6% false alarm rate.

Undergraduate Research Assistant – UM Soccer Analytics — May 2026 – Present
University of Michigan, SURE Program — *Advisor: Prof. Albert S. Berahas*

- Designed interpretable player importance scores for the UM men's and women's teams from 2025 Wyscout data, combining availability, share of team actions, and position-adjusted per-90 quality.
- Selected key drivers of match goal difference with Elastic Net, refit OLS models, and decomposed explained variance using partial R^2 and LMG shares.
- Built distribution-based win/draw/loss threshold analysis and expected probability curves linking team metrics to match outcomes, supporting coaching staff in match preparation and training priorities.
- Co-developed and currently maintain the UM Soccer analytics website.

Presentations

Y. Wang, M. Ellies, W. Long, A. Alsaidy, et al., *Propelling UM Soccer Using Data Analytics*. Poster, IOE Undergraduate Research Symposium, University of Michigan, Fall 2026.

Y. Wang and A. S. Berahas, *Data-Driven Insights for UM Soccer: Teams, Players, and Match Outcomes*. Poster, Summer Undergraduate Research in Engineering Symposium, University of Michigan, Summer 2026. [Poster]

Selected Projects

Tuning the Step: Wolfe Line-Search Parameter Sensitivity | *Optimization, Wolfe Line Search* — 2026

- Co-developed and benchmarked 10 continuous optimization algorithms on 12 test problems, including gradient descent, modified Newton, BFGS/DFP, and trust-region Newton-CG/SR1-CG methods.
- Compared convergence using optimality gaps, iteration counts, function and gradient evaluations, and CPU time. Examined how problem conditioning and curvature affected algorithm performance.
- Tested 90 Wolfe line-search parameter pairs (c_1, c_2) across four algorithms, using heatmaps and performance profiles to compare convergence speed and robustness.
- Conducted local sensitivity analysis by sampling 100 nearby parameter configurations around each selected pair, measuring changes in function evaluations under small perturbations.
- Selected modified Newton with Wolfe line search at $(c_1, c_2) = (0.05, 0.5)$ for its balance of speed and robustness.

Ride Smartly: Hourly Bike-Sharing Demand Prediction | *Time Series, XGBoost, MLP–LSTM* — 2026

- Forecast hourly bike rental demand for Washington, D.C.'s Capital Bikeshare system.
- Designed features and contributed to a hybrid model that adds a residual LSTM over 72-hour sequences on top of the MLP, refining predictions around demand peaks and valleys (final test $R^2 = 0.965$).

Enhanced Diffusion Colorizer | *Python, PyTorch, CNNs, U-Net, Diffusion Models* 2025

- Developed a two-stage automatic image colorization framework based on the pre-trained SIGGRAPH17 model.
- Designed an Enhanced U-Net integrating diffusion model, residual correction, global color priors, and decoder self-attention.
- Achieved improved performance: MSE 63.812, PSNR 30.500, SSIM 0.983, representing a 19.03% MSE improvement over SIGGRAPH17.
- Produced more vivid and semantically consistent colorizations than the baseline on FFHQ, and qualitatively tested generalization on out-of-domain images such as multi-face photos and natural scenes.

Fatal Police Shootings Statistical Analysis | *Statistical Modeling, Hypothesis Testing, Poisson Distribution* 2025

- Analyzed 2015–2025 U.S. fatal police shooting records with chi-squared goodness-of-fit and independence tests, showing daily counts follow a Poisson distribution ($\hat{\lambda} \approx 2.85$) with a significant shift during COVID-19.
- Found significant weekday and month dependence, and constructed confidence and prediction intervals to forecast 2025 shooting counts.

Soul Symphony | *Game Development, Rhythm Game* 2024

- Developed a 2D platform game that combines music, interactive gameplay, and storytelling.
- Implemented interactive platforms, rhythm controls, and level progression.

Mobius | *2D Game Development* 2024

- Developed a 2D brick breaking game with custom collision detection and multiple levels.
- Designed game levels and tested collision handling and gameplay mechanics.

Positions

Teaching Assistant May 2025 – Aug. 2025

Shanghai Jiao Tong University, Global College

- PHYS1410J Physics Lab (Simple Harmonic Motion); Instructor: Wang Yuxing.
- Led laboratory sessions, graded lab reports, and held office hours to support students.
- MATH2140J Linear Algebra; Instructor: Álvaro Mateos González.
- Led recitation sessions, graded homework and exams, and held office hours to assist students.

SilverFOCS Summer Intern May 2024 – Aug. 2024

FOCS Company

- Collaborated with a team to develop two games, contributing to gameplay mechanics, testing, and level design.

Network and Information Management Staff Sept. 2023 – Aug. 2025

Shanghai Jiao Tong University, Network and Information Center

- Maintained wireless and wired networks in campus dormitories.
- Responded to students' requests, troubleshot network issues, and ensured network stability.

International Program Office Assistant May 2025 – Aug. 2025

Shanghai Jiao Tong University

- Welcomed and assisted incoming exchange students from partner institutions.
- Provided logistical support and campus orientation.

Honors and Awards

University of Michigan: Dean's List and University Honors — Fall 2025 and Winter 2026.

Shanghai Jiao Tong University: Academic Excellence Scholarship — Class B (2024), Class C (2025).

Technical Skills

Programming Languages: Python, C/C++, MATLAB, SQL, Elm
Machine Learning & Data Science: PyTorch, scikit-learn, XGBoost, NumPy, pandas, Matplotlib
Developer Tools: Git, Docker, Linux, Slurm, VS Code, LaTeX
Technical Areas: Optimization, Statistical Modeling, Computer Vision, Time Series Analysis
Languages: Mandarin (Native), English (Fluent), German (Basic)